DAC Testnet

DAC Wallet Intelligence Layer

Wallet Intelligence, Dynamic Wallet Status SBT, and DAC Sender Launchpad Integration

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Environment	DAC Quantum Chain Testnet · GitHub Pages · Browser Frontend · DAC Explorer + RPC
Report Type	Application Development Report — Wallet Intelligence & Dynamic SBT Tooling
Version Scope	Wallet Intelligence Layer v1 (up to v1.5.4) · Wallet Intelligence Layer v2 (up to v2.0.2)

Community-built engineering contribution for DAC Testnet analytics and infrastructure testing.

Executive Summary

This report documents the design, development, and completion of DAC Wallet Intelligence Layer — a client-side wallet intelligence interface and dynamic wallet-bound badge layer for the DAC Quantum Chain Testnet. The work was completed across two development tracks: Wallet Intelligence Layer v1 (up to v1.5.4) and Wallet Intelligence Layer v2 (up to v2.0.2). v1 focused on public-data wallet analysis; v2 extended that foundation into a dynamic Wallet Status SBT system integrated with the DAC Sender NFT Launchpad.

This contribution follows the earlier DAC SENDER tooling work (Report 6, May 22 2026). DAC SENDER was designed to generate diverse measurable testnet activity; Wallet Intelligence Layer was designed to observe, classify, and document wallet behavior from public DAC Testnet data.

Component / Finding	Status
Wallet Intelligence Layer v1 (up to v1.5.4)	**Completed and documented**
Wallet Intelligence Layer v2 (up to v2.0.2)	**Completed and documented**
Wallet Status SBT contract	**Deployed and operational**
Dynamic SVG badge preview	**Confirmed in WIL v2 frontend**
DAC Sender NFT Launchpad integration	**Completed**
Deprecated development registry entries	**Hidden from Launchpad UI**
DAC / Truebit Etherscan API task-library PR	**Description updated**

Live Interfaces & Repository Links

Item	Live Interface	Repository Folder
Wallet Intelligence Layer v1	edlweiss186.github.io/.../v1/	GitHub — wallet-intelligence-layer-v1

Wallet Intelligence Layer v2	edlweiss186.github.io/.../v2/	GitHub — wallet-intelligence-layer-v2
DAC / Truebit PR	truebit-etherscan-api-task-library / PR #2	—

Project Context and Development Continuity

DAC Wallet Intelligence Layer was developed as a continuation of the DAC SENDER testnet tooling contribution. DAC SENDER produced diverse EVM transaction patterns across native transfers, contract deployments, NFT deployment, NFT minting, and registry interactions. Wallet Intelligence Layer adds the complementary observation layer — reading public DAC Testnet activity and converting it into structured wallet intelligence.

```
DAC SENDER
  → generates diverse DAC Testnet activity

Wallet Intelligence Layer
  → observes, classifies, and documents wallet activity
```

The concept also connects to the DAC / Truebit Etherscan API task-library work, specifically the dac_wallet_intelligence function-task direction. The PR description in the related DAC repository was updated to reference the live Wallet Intelligence Layer v1 and v2 implementations.

System Overview

Wallet Intelligence Layer is implemented as a static, no-backend frontend. The checker does not require wallet connection for analysis. Users paste a wallet address and the frontend reads public explorer and RPC data to produce a structured wallet profile. In v2, the same interface can optionally trigger a wallet transaction to mint or update a dynamic SBT.

Application Type	Static HTML / CSS / JavaScript frontend
Hosting	GitHub Pages
Build Step	None
Backend	None
Primary Data Sources	DAC Explorer API + DAC Testnet RPC
Wallet Check Model	Pasted-address check — no wallet connection required
Mint / Update Model	User-triggered smart contract transaction via browser wallet popup

User Interaction Model

Component / Finding	Status
Paste wallet address and run check	**No wallet connection; no signature required**
View wallet intelligence output	**Read-only frontend output**

Preview dynamic badge	**Read-only contract / frontend render**
Mint Wallet Status SBT	**Browser wallet popup — user signs transaction**
Update Wallet Status SBT	**Browser wallet popup — user signs transaction**

Network and Data Configuration

Network	DAC Quantum Chain Testnet
Chain ID	21894 (0x5586)
Native Token	DACC
RPC Endpoint	https://rpctest.dachain.tech/
Explorer	https://exptest.dachain.tech
Explorer API	https://exptest.dachain.tech/api
DAC Inception Rank Contract	0xB36ab4c2Bd6aCfC36e9D6c53F39F4301901Bd647
Wallet Status SBT Contract	0xdE02dfbf28563f80733423678477357dF9040b14
DACNFTRegistry	0x34CDf37FeEb81877acabAD601AAaA3AE5a5029Ae

Wallet Intelligence Layer v1 — v1.5.4

Wallet Intelligence Layer v1 is a read-only DAC Testnet wallet checker. It takes a pasted wallet address and converts public explorer and RPC data into a structured wallet profile. The final v1 version is v1.5.4.

Core Modules

- Proof of Native Funds
- Proof of Assets Engine
- Activity Analytics
- NFT Portfolio Intelligence
- Reputation Scoring
- Sybil Heuristics
- DAC Inception Rank signal
- DACC stake signal
- Wallet Quality Policy display
- Stress testing links
- Raw JSON wallet profile output
- Read Project Details / GitHub project button (added in v1.5.4)

v1.5.4 Status

Component / Finding	Status
Read-only pasted-address wallet checking	**Implemented**
No wallet connection requirement	**Confirmed**
Explorer / RPC public-data profile generation	**Implemented**
Wallet scoring and policy display	**Implemented**
GitHub project folder link	**Added in v1.5.4**

Wallet Intelligence Layer v2 — v2.0.2

Wallet Intelligence Layer v2 extends the v1 wallet intelligence model with a dynamic wallet-bound badge layer. The wallet checker remains a client-side pasted-address flow, while minting or updating the badge is an optional wallet-signed transaction.

Functional Scope

- Dynamic Wallet Status SBT preview generated from the checked wallet profile
- One evolving badge per wallet
- ERC-5192-style locked / non-transferable SBT behavior
- Dynamic tokenURI metadata
- Dynamic SVG image rendering per wallet
- DAC Inception Rank signal integration
- Native DACC balance tier signal
- Wallet archetype classification
- Mint / update transaction trigger from frontend
- DAC Sender NFT Launchpad registry integration
- Dynamic artwork support inside DAC Sender NFT Launchpad

NFT Description

Wallet Status is a dynamic DAC Testnet badge generated by the DAC Wallet Intelligence Layer v2. Each wallet can hold one evolving badge that reflects its verified core on-chain status, DAC Inception Rank signal, native DACC balance tier, and wallet profile state.

Wallet Status SBT Architecture

Wallet Status SBT is the on-chain badge representation of the wallet profile. The contract is designed around one wallet holding one locked badge. Transfers and approvals are disabled to preserve wallet-status integrity.

Contract	0xdE02dfbf28563f80733423678477357dF9040b14
Name	Wallet Status SBT
Symbol	Status

Token Type	ERC-5192 locked SBT
Mint Price	1 DACC
Update Price	0.001 DACC
Max Mint per Wallet	1
Rank Badge Source	0xB36ab4c2Bd6aCfC36e9D6c53F39F4301901Bd647

Contract Interface

Launchpad compatibility functions:

```
name()  ·  symbol()  ·  totalSupply()  ·  maxSupply()
maxMintPerWallet()  ·  mintPrice()  ·  mintedPerWallet(address)  ·  mint(uint256)
```

Dynamic badge helpers:

```
previewProfile(address)  ·  badgeTextOf(address)
dynamicSvgOf(address)  ·  dynamicImageOf(address)
tokenURI(uint256)  ·  needsUpdate(address)  ·  updateBadge()
```

SBT Transferability Model

```
locked(uint256)          → true
approve()                → revert
setApprovalForAll()      → revert
transferFrom()           → revert
safeTransferFrom()       → revert
```

Token Metadata

- Dynamic JSON metadata returned through tokenURI(tokenId)
- Dynamic SVG image tied to the wallet profile
- Traits: badge class, pattern health, Sybil risk, inception rank, archetype, rank badge count, balance tier, wallet address, token type, transferability, image mode, and verification mode

DAC Sender NFT Launchpad Integration

Wallet Status SBT is registered into the existing DAC Sender NFT Launchpad registry. The final collection appears in the Launchpad Collections tab, and minted badges appear in My Collections after a user mints the SBT.

Registry	0x34CDf37FeEb81877acabAD601AAaA3AE5a5029Ae
Registered Collection	Wallet Status SBT
Final Contract	0xdE02dfbf28563f807334236784773 57dF9040b14
Collection View	Visible under Collections tab
User View	Visible under My Collections after mint

Dynamic Artwork Compatibility

The final SBT uses dynamic wallet-specific SVG artwork. DAC Sender NFT Launchpad was extended to render dynamic SVG output from the Wallet Status SBT contract rather than relying only on static collection images.

```
connected wallet address
  → dynamicSvgOf(address)
  → inline SVG render in Launchpad
```

Deprecated Registry Entries

During development, two earlier Wallet Status contracts were registered while the dynamic badge workflow was still being validated. Those versions exposed issues including static artwork behavior and dynamic image rendering inconsistencies.

Hidden development entries:

```
0xC790a1f6b314267e88F6ef2FcB66537DD2C4e1a2
0x28aA8D67Bb4BeCF5ad6fc5E28f5B971FebB29cFf
```

Final visible version:

```
0xdE02dfbf28563f80733423678477357dF9040b14
```

The entries are not removed from the on-chain registry because the registry appears append-only from the available frontend interface. The Launchpad UI filters those known development entries so users only see the finalized Wallet Status SBT implementation.

DAC / Truebit Etherscan API Task-Library PR Update

The PR description for the related DAC / Truebit Etherscan API task-library contribution was updated to reference the completed Wallet Intelligence Layer implementation. This connects the original dac_wallet_intelligence task concept with the live frontend implementation and the follow-up dynamic SBT layer.

Pull Request	https://github.com/dacblockchain/truebit-etherscan-api-task-library/pull/2
v1 Reference	Wallet Intelligence Layer v1 — up to v1.5.4
v2 Reference	Wallet Intelligence Layer v2 — up to v2.0.2
Purpose	Connect task-library concept, live wallet checker, and dynamic SBT follow-up

Testing and Validation Results

Component / Finding	Status
Wallet Intelligence Layer v1 live interface	Confirmed
Wallet Intelligence Layer v2 live interface	Confirmed
No-connect wallet check model	Confirmed

Dynamic badge preview in WIL v2	**Confirmed**
Wallet Status SBT final contract deployment	**Confirmed**
Mint flow from WIL v2	**Confirmed by user testing**
Explorer visibility after mint	**Confirmed**
Launchpad Collections visibility	**Confirmed**
Launchpad My Collections visibility after mint	**Confirmed**
Dynamic artwork rendering in Launchpad	**Patched and confirmed**
Old Wallet Status registry entries hidden	**Confirmed**

Technical Findings During v2 Finalization

- Static IPFS artwork was insufficient for wallet-specific SBT output — each wallet requires a unique dynamic badge view.
- SVG path recreation of the DAC logo was visually unreliable; using the original IPFS logo asset produced a better final result.
- Rendering external IPFS assets inside a data:image/svg+xml image context caused broken-image behavior in some browser contexts.
- Inline SVG rendering resolved the dynamic preview issue for Wallet Intelligence Layer and Launchpad views.
- Registry entries created during iterative development needed to be filtered from the Launchpad UI because the registry appears append-only.
- The final user experience required checking the web preview before minting or registering further changes.

Security Model and Limitations

Wallet Check Security

- No wallet connection is required for checking a wallet.
- No transaction signature is required for analysis.
- No private key access is possible through the checker.
- No account login or backend profile store is used.
- Only public explorer and RPC data is read.

Mint / Update Security

- Mint and update actions are triggered only by explicit user action.
- The browser wallet popup is required before any transaction can be submitted.
- The user manually reviews and signs the transaction.
- Private keys remain inside the wallet client.
- The frontend does not custody funds or tokens.

Limitations

- Wallet classifications are community-defined engineering interpretations, not official DAC determinations.
- The system does not prove whether a wallet is Sybil or non-Sybil.
- The SBT is a testnet identity and analytics experiment, not a financial score or official eligibility badge.
- Dynamic artwork depends on contract reads, RPC availability, and IPFS asset availability.
- Launchpad filtering hides deprecated development entries at the UI layer rather than deleting registry state.

Recommendations

For DAC Team / Ecosystem Review

1. Consider native Launchpad support for dynamic token artwork, especially SVG generated from contract reads.
2. Consider exposing registry management functions or an official deprecation flag if registry entries are expected to evolve over time.
3. Continue improving RPC reliability — wallet intelligence and launchpad rendering rely on consistent read access.
4. Consider publishing official guidance for testnet contribution tools that read explorer and RPC data to generate wallet analytics.

For Future Wallet Intelligence Layer Development

5. Add badge evolution history and status update events.
6. Add a dedicated event history panel for Wallet Status SBT updates.
7. Expand known DAC ecosystem collection detection.
8. Improve historical activity visualization and wallet behavior timelines.
9. Document additional wallet archetypes as more testnet behavior patterns become observable.

Conclusion

DAC Wallet Intelligence Layer has completed the transition from concept to working testnet tooling. v1 provides the read-only wallet intelligence interface, while v2 extends that foundation into a dynamic wallet-bound SBT system integrated with the existing DAC Sender NFT Launchpad registry.

The final system demonstrates a complete client-side workflow: paste a wallet address, inspect the wallet profile, preview a wallet-specific badge, optionally mint or update a locked SBT, and view the result through the DAC Sender Launchpad and explorer. This creates a practical observation layer for DAC Testnet activity without introducing backend custody or account-based infrastructure.

Component / Finding	Status
Wallet Intelligence Layer v1.5.4	**Completed**
Wallet Intelligence Layer v2.0.2	**Completed**
Wallet Status SBT	**Deployed and operational — 0xdE02dfbf28563f80733423678477357dF9040b14**

Launchpad integration	**Completed**
PR description update	**Completed**